

Financial Accounting

Executive MBA

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Lecture 7: Income Tax Reporting Issues



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Topics

- Understanding the difference between Book Income (income before taxes reported for GAAP purposes) vs. Taxable Income (income reported to a tax authority).
- Reconciling the difference between income tax expense for a period vs. taxes owed (i.e., increase in taxes payable) for the same period.
- Understanding Tax Disclosures from financial reports.

Income Tax Reporting: Terminology

- **Book income** is income measured using GAAP (US GAAP or IFRS) during a fiscal period.
 - It equals GAAP revenues less GAAP expenses.
- **Taxable income** is the income measured using the tax code of a jurisdiction.
 - It equals taxable revenues less tax-deductible expenses that are measured by the tax rules in force.
- Book income *differs* from taxable income for two reasons:
 1. Temporary book-tax differences.
 2. Permanent book-tax differences.

Income Tax Reporting: Terminology cont'd

- **Income tax expense or provision for income taxes** is a GAAP operating expense recognized on the income statement during a fiscal period.
 - $\text{Book Income} - \text{Income Tax Expense} = \text{Net Income}$
- **Income tax payable (receivable)** is a liability (asset) that captures taxes owed to (due from) tax authorities (federal, states, cities, and foreign) at any point in time.
- **Deferred taxes** are noncurrent assets (deferred tax assets) or noncurrent liabilities (deferred tax liabilities) that capture *temporary book-tax differences* at any point in time.

Tax Terminology cont'd

- **Statutory tax rate (STR):** Set by tax laws of a country (or jurisdiction).
 - $\text{Taxes payable} = \text{Taxable income} \times \text{statutory tax rate}$.
- **Effective tax rate (ETR):** Calculated as the ratio of income tax expense to book income and therefore derived using GAAP numbers. It captures the amount of taxes a company will end up paying on each dollar of GAAP earnings.
- Statutory tax rate differs from effective tax rate for two main reasons:
 - a. Permanent book-tax differences. *We will focus on this difference.*
 - b. Foreign, state, and local taxes.

GAAP (Book) Income Statement vs. Tax Return

GAAP Income Statement		Tax Return
GAAP Revenues	$\neq$	Taxable Revenues
– GAAP Expenses	$\neq$	– Tax Deductible Expenses
= Book (GAAP) Income	$\neq$	= Taxable Income
– Income Tax Expense ($\neq$ Book income $\times$ STR).	$\neq$	$\times$ STR
Net Income		= Taxes Owed (Increase in taxes payable)

- We will learn that income tax expense during a period is rarely equal to taxes owed during that same period. The difference is reconciled by changes in deferred tax assets and/or liabilities.
- We will also learn that Income Tax Expense $\neq$ Book Income $\times$ STR. This is due permanent book-tax differences.

Temporary Book-Tax Differences

- A **temporary difference** arises when a revenue item or expense item affects book income in one period but affects taxable income in a *different (either earlier or later) period*.
 - Temporary differences arising in the current period that cause taxable income to be *higher* than book income in *future* periods give rise to future taxable amounts called **Deferred Tax Liabilities (DTL)**.
 - Conversely, temporary differences arising in the current period that cause taxable income to be *lower* than book income in *future* periods give rise to future deductible amounts called **Deferred Tax Assets (DTA)**.
- A temporary difference results in a difference between the book value (amount recognized on the **GAAP balance sheet**) and tax basis (amount recognized on the **tax balance sheet**) of an asset or a liability that will result in **taxable amounts** or **deductible amounts** in future years.

Four Types of Temporary Differences

1. Revenues/gains recognized for GAAP *before* recognized for tax:
 - Unrealized gains for marking to market assets for GAAP purposes but such gains are only recognized for tax purposes when they are realized, i.e., when assets are sold.
2. Expenses/losses recognized for GAAP *before* recognized for tax:
 - Product warranty liabilities resulting in warranty expense for GAAP purposes that are tax deductible in the future when actual warranty expenditures are made.
 - Bad debt expense recognized using the allowance method for GAAP purposes that are tax deductible in the future when receivables are written off.
3. Revenues/gains recognized for tax *before* recognized for GAAP :
 - Subscriptions received in advance.
 - Cash receipts from Advance Orders.
4. Expenses/losses recognized for tax *before* recognized for GAAP:
 - Depreciable property.
 - Prepaid expenses that are deducted on the tax return in the period paid.

Permanent Differences

- **Permanent differences** result from items that either (1) enter into book income but **never** into taxable income or (2) enter into taxable income but **never** into book income.
 - Permanent differences affect only the period in which they occur. Therefore, *unlike temporary differences*, they do not give rise to future taxable or deductible amounts resulting in **no recognition of deferred tax assets or liabilities**.
- Examples of permanent differences that affect book income but never taxable income:
 - a. Non-deductible penalties and fines from violations of the law.
 - b. Non-taxable interest income on state/municipal securities in the United States.
 - c. Book Income from foreign subsidiaries that are permanently reinvested in foreign subsidiaries and hence never affect taxable income.
- Examples of permanent difference that affect taxable income but not book income:
 - a. Excess tax benefits from stock-based compensation that reduce taxable income but not book income.
 - b. Tax credits that reduce taxes owed and therefore income tax expense but not book income.

Book-Tax Differences: Example 1

- Consider a firm that is incorporated with the following identical book and tax balance sheet at the beginning of fiscal year 1 (FY1):

Balance Sheet (Book and Tax)	Beginning of FY1
Cash	\$1,000
Equipment	<u>900</u>
Total Assets	<u>\$1,900</u>
Total Equity	<u>\$1,900</u>

- The firm depreciates the equipment over 3 years by the straight-line method for GAAP purposes but over 2 years for tax purposes. For Example 1, we will assume that depreciation expense constitutes the *only* book-tax difference so that book income *before* depreciation = taxable income *before* depreciation = ***cash income before taxes*** of \$1,000 for the next 3 fiscal years.
- The statutory tax rate is expected to be 30% for the future and any taxes owed during a fiscal year will be paid in cash at the end of the fiscal year.

Example 1 cont'd

	(a)	(b)
Fiscal Year (FY)	Book Income (\$1,000 – Book Depreciation)	Taxable Income (\$1,000 – Tax Depreciation)
FY1	\$700	\$ 550
FY2	700	550
FY3	700	1,000
Cumulative	\$2,100	\$ 2,100

Fiscal Year (FY)	(c) Book income – Taxable Income	(d) Taxes owed [Col (b) x 30%] (current taxes)	(e) Increase (Decrease) in Deferred Tax Liability [Col (c) x 30%]	(f) Income Tax Expense (Current expense +/- Deferred expense) [Col (d) + Col (e)]
FY1	$\$700 - 550 = \150	\$165	\$45	\$210
FY2	$700 - 550 = 150$	165	45	210
FY3	$700 - 1,000 = -300$	300	-90	210
Cumulative	\$0	\$630	\$0	\$630

Example 1: Tax Entries

FY1		
Income Tax Expense [derived]	$\$210 = \$45 + 165$	
	Deferred Tax Liability (DTL) [Increase in DTL = $45 - 0$]	45
	Cash taxes (calculated) [Taxable income of $\$550 \times 30\%$]	\$165
FY2		
Income Tax Expense [derived]	$\$210 = \$45 + 165$	
	Deferred Tax Liability (DTL) [Increase in DTL = $90 - 45$]	45
	Cash (calculated) [Taxable income of $\$550 \times 30\%$]	\$165
FY3		
Income Tax Expense [derived]	$\$210 = \$45 + 165$	
Deferred Tax Liability [Decrease in DTL = $0 - 90$]	90	
	Cash (calculated) [Taxable income of $\$1,000 \times 30\%$]	\$300

Example 1: Financial Statements

Income Statements		FY1		FY2		FY3
Income before Taxes		\$700		\$700		\$700
– Income Tax Expense						
Current	165		165		300	
+/- Deferred	45	–210	45	–210	–90	–210
Net Income		\$490		\$490		\$490

Tax Returns	FY1	FY2	FY3
Taxable Income	\$550	\$550	\$1,000
x Statutory tax rate	30%	30%	30%
= Taxes owed	\$165	\$165	\$300

Derive the firm's effective tax rates for FY1–3. Reconcile those rates with the firm's statutory tax rates.

Example 1: Financial Statements continued...

Statement of Cash Flows	FY1	FY2	FY3
Net Income	\$ 490	\$ 490	\$ 490
+ Depreciation Expense	300	300	300
+ (–) Increase (Decrease) in DTL	45	45	–90
CFFO	\$835	\$835	\$700

- Note that we have assumed that any taxes owed are paid in cash at the end of each fiscal year.

Complete the schedule below to capture the scenario in which *taxes owed were instead accrued at the end of each fiscal year and paid in cash in the following fiscal year?*

Statement of Cash Flows	FY1	FY2	FY3
Net Income	\$ 490	\$ 490	\$ 490
+ Depreciation Expense	300	300	300
+/(–) Increase (Decrease) in DTL	45	45	–90
+/(–) Increase (Decrease) in Taxes Payable			
CFFO			

Example 1: GAAP vs. Tax Balance Sheets

GAAP Balance Sheets	FYE0	FYE1	FYE2	FYE3
Cash	\$1,000	\$1,835	\$2,670	\$3,370
PP&E-Net	900	600	300	0
Total Assets	\$1,900	\$2,435	\$2,970	\$3,370
Deferred Tax Liability (Noncurrent)	0	45	90	0
Equity	1,900	2,390	2,880	3,370
Total Liabilities & Equity	\$1,900	\$2,435	\$2,970	\$3,370

Tax Balance Sheets	FYE0	FYE1	FYE2	FYE3
Cash	\$1,000	\$1,835	\$2,670	\$3,370
PP&E-Net	900	450	0	0
Total Assets	\$1,900	\$2,285	\$2,670	\$3,370
Equity	1,900	2,285	2,670	3,370
Total Liabilities & Equity	\$1,900	\$2,285	\$2,670	\$3,370

Takeaways: Example 1

- In any given fiscal year, *tax expense does not equal taxes owed (increase in taxable payable)*, requiring the use of balance sheet accounts, i.e., changes in deferred tax assets and deferred tax liabilities to reconcile the difference:

Income tax expense = Increase in Taxes Payable +/- changes in deferred tax assets and liabilities.

- These DTAs/DTLs, in turn, ensure that the effective tax rate of the company remains the same as the statutory tax rate:
 - Therefore, income tax expense recognized in a fiscal year is based on the income tax consequences of the current period's pretax income, regardless of when those income tax consequences are realized in cash.

Takeaways: Example 1 cont'd

- Temporary differences result in the book (GAAP) values of assets and liabilities to differ from their tax bases resulting in deferred tax assets/liabilities recognized on the book balance sheets.
- Note that for Example 1, the DTL on the balance sheet (Slide 15) at any point in time can be calculated as follows:

$$DTL = [PP\&E \text{ at book value} - PP\&E \text{ at tax value}] * STR$$

- In general, DTAs and DTLs imply that the book values of assets and liabilities differ from their tax bases due to recognition of revenues and/or expenses in different periods for book purposes vs. tax purposes.

Example 2

- Example 2 is similar to Example 1 except for *two additional book-tax differences*:
 1. During FY1, the firm was fined \$100 for engaging in illegal transactions on the island of Mauritius. It accrued a GAAP liability of \$100 during that year. The fine will be paid in cash in FY2.
 - The fine is disallowed as a tax deduction (even when it is paid), so the fine liability results in a *permanent book-tax difference* in FY1.
 2. The firm received advance cash of \$200 from a customer in FY1 to deliver services worth \$200 in FY2. For tax purposes, cash collected in advance are taxable revenues.
 - The recognition of taxable revenues in FY1 is a *temporary difference* as those revenues are not recognized for GAAP purposes until FY2. Therefore, current taxable income exceeds current book income (resulting in a *deferred tax asset* (DTA) created in FY1) but future taxable income would be *lower* than future book income (resulting in a reversal of the DTA in FY2).

Example 2 continued...

Income Statement (before income tax expense)	FY1
Book Income before Book-Tax Differences	\$1,000
– Depreciation Expense	–300
– Fine	–100
= Book Income	\$ 600

Tax Return	FY1
Taxable Income before Book-Tax Differences	\$1,000
+ Revenues from Advance Cash Orders	200
– Depreciation Expense	–450
= Taxable Income	\$ 750

Example 2 continued...

- As in Example 1, excess of book over taxable income due to depreciation in FY1 results in a deferred tax liability (DTL) of \$45.
- However, taxable income is **higher than** book income by \$200 in FY1 due to the advance cash orders resulting in the creation of a deferred tax asset (DTA) of $\$200 \times 30\% = \60 .
- The fine of \$100 accrued in FY1 for book purposes will never be tax deductible. It is therefore a permanent tax difference resulting in no deferred tax items.
- Taxes owed in FY1 = taxable income $\times 30\% = \$750 \times 30\% = \225 .
- Tax Entry for FY1:

$$\text{Income tax expense} = \text{Taxes owed} + \text{increase in DTL} - \text{increase in DTA}$$

FY1		
Income Tax Expense [derived]	\$210	
Deferred Tax Asset (calculated)	60	
	Deferred Tax Liability (calculated)	\$ 45
	Cash (calculated) [Taxable income of \$750 x 30%]	\$ 225

Example 2 cont'd

Income Statement		FY1
Income before Book-Tax Differences		\$1,000
–Depreciation Expense		–300
–Fine		–100
= Income before Taxes		\$ 600
– Income tax expense		
Current	225	
+/- Deferred, net	–15	–210
= Net Income		\$ 390

Effective Tax Rate vs. Statutory Tax Rate

- Calculate the effective tax rate for FY1:
- Using the schedule below, reconcile the difference between the statutory tax rate of 30% and the effective tax rate for FY1:

Effective tax rate for FY1 if there were no permanent book-tax differences (equivalently, all book income were taxed at the STR)	30%
+ Increase in effective tax rate due to non-deductible fine	
Effective Tax Rate for FY1	

Example 2 continued...

Income Statement (before income tax expense)	FY2
Book Income before Book-Tax Differences	\$1,000
+ Reversal of Deferred Revenue	200
– Depreciation Expense	–300
= Book Income	\$ 900

Tax Return	FY2
Taxable Income before book-tax differences	\$1,000
– Depreciation Expense	–450
= Taxable Income	\$550

Example 2 continued...

- Excess of tax depreciation over book depreciation results in an increase in a deferred tax liability of \$45 in FY2.
- However, in FY2, taxable revenues are *lower* than book revenues by \$200 as services are rendered for advance cash orders. This, in turn, results in a full reversal (decrease) of the deferred tax asset of \$60 set up in FY1.
- Taxes paid in FY2 = taxable income*30% = \$550*30% = \$165.
- Tax Entry for FY2:

Income tax expense = taxes payable + increase in DTL + decrease of DTA

FY2		
Income Tax Expense [derived]	\$270	
	Deferred Tax Asset (calculated)	\$ 60
	Deferred Tax Liability (calculated)	\$ 45
	Cash (calculated) [Taxable income of \$550 x 30%]	\$ 165

Example 2

Income Statement		FY2
Income before Book-Tax Differences		\$1,000
Depreciation Expense		–300
Reversal of Deferred Revenue		200
Income before Taxes		\$ 900
– Income tax expense		
Current	165	
+ Deferred	105	–270
Net Income		\$ 630

- *Derive the effective tax rate for FY2. How does it compare with the statutory tax rate of 30%.*

Example 2 continued...

Income Statement (before income tax expense)	FY3
Book Income before Book-Tax Differences	\$1,000
–Depreciation Expense	–300
= Income before Taxes	\$ 700

Tax Return	FY3
Taxable Income before book-tax differences	\$1,000
–Depreciation Expense	–0
= Taxable Income	\$1,000

Example 2 continued...

FY3		
Income Tax Expense [derived]	\$210	
Deferred Tax Liability	90	
	Cash (calculated) [Taxable income of \$1,000 x 30%]	\$ 300

Income Statement		FY3
Income before Book-Tax Differences		\$1,000
–Depreciation Expense		–300
Income before Taxes		\$ 700
– Income tax expense		
Current	300	
+/- Deferred	–90	–210
Net Income		\$ 490

Example 2: Financial Statements continued...

Statement of Cash Flows	FY1	FY2	FY3
Net Income	\$ 390	\$ 630	\$ 490
+ Depreciation Expense	300	300	300
+/(−) Increase (Decrease) in DTL	45	45	−90
+/(−) Decrease (Increase) in DTA	−60	60	0
+/(−) Increase (Decrease) in Fine Payable	100	(100)	0
+/(−) Increase (Decrease) in Deferred Revenues	200	(200)	0
Cash flows from operations (CFFO)	\$975	\$735	\$700

In practice, these two items are combined on the cash flow statement and disclosed as "change in net deferred taxes."

Statement of Cash Flows	FY1	FY2	FY3
Net Income	\$ 390	\$ 630	\$ 490
+ Depreciation Expense	300	300	300
+/(−) Change in Net Deferred Taxes	−15	105	−90
+/(−) Increase (Decrease) In Fine Payable	100	(100)	0
+/(−) Increase (Decrease) in Deferred Revenues	200	(200)	0
Cash flows from operations (CFFO)	\$975	\$735	\$700

Example 2: Book vs. Tax Balance Sheets

GAAP Balance Sheet	FYE0	FYE1	FYE2	FYE3
Cash	\$1,000	\$1,975	\$2,710	\$3,410
Equipment	900	600	300	0
DTA-net of DTL	0	15	0	0
Total Assets	\$1,900	\$2,590	\$3,010	\$3,410
DTL-net of DTA	0		90	0
Fine Payable	0	100	0	0
Deferred Revenue	0	200	0	0
Equity	1,900	2,290	2,920	\$3,410
Total Liabilities & Equity	\$1,900	\$2,590	\$3,010	\$3,410
Tax Balance Sheet	FYE0	FYE1	FYE2	FYE3
Cash	\$1,000	\$1,975	\$2,710	\$3,410
Equipment	900	450	0	0
Total Assets	\$1,900	\$2,425	\$2,710	\$3,410
Equity	1,900	2,425	2,710	\$3,410
Total Liabilities & Equity	\$1,900	\$2,425	\$2,710	\$3,410

Example 2 Takeaways

- Both temporary and permanent differences cause the book balance sheet to differ from the tax balance sheet. However, only temporary differences create deferred tax assets or liabilities (deferred taxes) on the book balance sheet. Permanent differences do not.

➤ At FYE1, net DTA of \$15 on the balance sheet arises as follows:

$$DTL_{PPE} = [600 - 450] * 0.30 = \$45$$

$$DTA_{Def Rev} = [200 - 0] * 0.30 = \$60$$

➤ At FYE2, net DTL of \$90 on the balance sheet arises as follows:

$$DTL_{PPE} = [300 - 0] * 0.30 = \$90$$

Example 2: Takeaways

- However, it is permanent *not* temporary differences that cause the effective tax rate (ETR) to differ from the statutory tax rate (STR).
 - ETR = STR in all the fiscal years in Examples 1 and 2 *except for FY1 in Example 2*, when ETR differs from STR due to the non-deductible fine which is a permanent tax difference.
- Regardless of the nature of book-tax differences, income tax expense can still be derived as follows:

$$\begin{array}{ccccc} & & \textit{Change In} & & \\ & & \textit{Deferred} & & \\ & & \textit{Income Taxes} & & \\ & & \textit{arising from a} & & \\ & & \textit{firm's} & & \\ & & \textit{operations.} & & \\ \textit{Income Taxes} & +/ - & & = & \textit{Income Tax} \\ \textit{Payable} & & & & \textit{Expense} \end{array}$$

Example 2: Takeaways

- Alternatively, in the presence of permanent tax differences, income tax expense is calculated in two steps:

Step 1: Derive “Adjusted Book Income” as follows:

Adjusted Book income = Book income + non-deductible expenses – non-taxable revenues – excess tax benefits (tax deductions claimed on tax return but not on book income statement).

*Step 2: Income tax Expense = Adjusted book income * statutory tax rate.*

- *Therefore, income tax expense is only based on the portion of book income that affects taxable income regardless of when the taxes are paid or received.*

Presentation and Classification: Deferred Tax Items

- *For an individual item*, temporary difference always reverses eventually, and deferred tax asset or liability becomes zero.
- *For company as a whole*, there are always new items giving rise to temporary differences, so deferred tax asset or liability does not go necessarily to zero.
- Classification of DTAs and DTLs on the balance sheet:
 - All DTAs and DTLs classified as noncurrent assets and liabilities, respectively.
 - DTAs and DTLs may be netted against each other within a tax jurisdiction *but not* across different jurisdictions.

Deferred Taxes and Tax Rate Changes

- US GAAP and IFRS require companies to measure deferred tax assets and liabilities using enacted tax rates that will apply in the years in which the temporary differences are expected to be recovered or paid. So, what happens if tax rate changes in future years?
- Changes in tax rates are treated as *changes in accounting estimates*.
 - If tax rate changes, revalue DTAs and DTLs, affecting tax provision (income tax expense) in year the rate change is enacted.
- In 2017, New Tax Reform Act in the US reduced corporate tax rates from 35% to 21%:
 - Companies revalued all their DTAs and DTLs set up at the tax rate of 35% to the lower tax rate of 21%.
 - Such revaluations at the lower tax rate resulted in increases or decreases in tax expense.

Revaluation of Deferred Tax Items: US Tax Reform Act of 2017

Deutsche Bank has warned that it will record its third consecutive annual loss in 2017 because of a €1.5bn hit from US tax reforms and a 22 per cent fall in trading and financing revenues in the fourth quarter.

Germany's biggest lender, which had hoped to return to a full-year profit after pre-tax losses of almost €7bn in total for the two previous years, said on Friday that it would instead make a "small full-year, after-tax loss" for 2017.

The warning, which wiped 4 per cent off Deutsche's shares in late trading, came hours after Morgan Stanley joined a chorus of companies detailing one-off hits from the US tax reforms.

Morgan Stanley said its fourth-quarter earnings would be weakened by a \$1.25bn charge, largely because its deferred tax assets (DTAs), which can be offset against future tax bills, are worth less because of sweeping changes to the US tax code signed into law by President Donald Trump a few days before Christmas.

Deutsche said a change in the value of its DTAs were the main reason it will make a loss for 2017, instead of the €1.3bn profit after tax expected by analysts the bank polled on Jan 3.

Tax Disclosures in Financial Statements

- What can we learn about taxes from a company's Income Statement, Balance Sheet, and Cash Flow Statement?
- What can we learn from a company's income tax disclosure note?
 1. Components of tax provision: current vs. deferred taxes.
 2. Schedule of tax rate reconciling statutory tax rate to effective tax rate.
 3. Schedule of deferred taxes.

Understanding Tax Disclosures: Microsoft

INCOME STATEMENTS

(In millions, except per share amounts)

Year Ended June 30,	2024
Revenue:	
Product	\$ 64,773
Service and other	180,349
Total revenue	245,122
Cost of revenue:	
Product	15,272
Service and other	58,842
Total cost of revenue	74,114
Gross margin	171,008
Research and development	29,510
Sales and marketing	24,456
General and administrative	7,609
Operating income	109,433
Other income (expense), net	(1,646)
Income before income taxes	107,787
Provision for income taxes	19,651
<i>(aka as Income tax Expense)</i>	
Net income	\$ 88,136

Microsoft: Tax Disclosures

Provision for Income Taxes

The components of the provision for income taxes were as follows:

(In millions)

Year Ended June 30,	2024
Current Taxes	
U.S. federal	\$ 12,165
U.S. state and local	2,366
Foreign	9,858
Current taxes	\$ 24,389
Deferred Taxes	
U.S. federal	\$ (4,791)
U.S. state and local	(379)
Foreign	432
Deferred taxes	\$ (4,738)
Provision for income taxes	\$ 19,651

U.S. and foreign components of income before income taxes were as follows:

(In millions)

Year Ended June 30,	2024
U.S.	\$ 62,886
Foreign	44,901
Income before income taxes	\$ 107,787

1. Show the journal entry Microsoft made during fiscal 2024 to provision for income taxes?
2. Calculate Microsoft's effective tax rate for fiscal 2024. Provide some reasons why it might be different from the statutory tax rate of 21%?

Microsoft's SCF

(In millions)

Year Ended June 30,	2024
Operations	
Net income	\$ 88,136
Adjustments to reconcile net income to net cash from operations:	
Depreciation, amortization, and other	22,287
Stock-based compensation expense	10,734
Net recognized losses (gains) on investments and derivatives	305
Deferred income taxes	(4,738)
Changes in operating assets and liabilities:	
Accounts receivable	(7,191)
Inventories	1,284
Other current assets	(1,648)
Other long-term assets	(6,817)
Accounts payable	3,545
Unearned revenue	5,348
Income taxes	1,687
Other current liabilities	4,867
Other long-term liabilities	749
Net cash from operations	118,548

3. Using the above information, to reconcile the difference between the amount of taxes provisioned during fiscal 2024 vs. the amount of income taxes paid in cash during fiscal 2024.

Microsoft: Reconciliation of Statutory vs. Effective Tax Rate

The items accounting for the difference between income taxes computed at the U.S. federal statutory rate and our effective rate were as follows:

Year Ended June 30,	2024
Federal statutory rate	21.0%
Effect of:	
Foreign earnings taxed at lower rates	(1.4)%
Impact of intangible property transfers	0%
Foreign-derived intangible income deduction	(1.1)%
State income taxes, net of federal benefit	1.5%
Research and development credit	(1.1)%
Excess tax benefits relating to stock-based compensation	(1.1)%
Interest, net	1.1%
Other reconciling items, net	(0.7)%
Effective rate	18.2%

4. The above schedule shows that during fiscal 2024, Microsoft's effective tax rate was higher than its statutory tax rate by 1.1% due the interest, net. What was amount of interest expense that was recognized for GAAP but disallowed for tax purposes during fiscal 2024?

Microsoft's Schedule of Deferred Tax Items

The components of the deferred income tax assets and liabilities were as follows:

(In millions)		
June 30,	2024	2023
Deferred Income Tax Assets		
Stock-based compensation expense	\$ 765	\$ 681
Accruals, reserves, and other expenses	4,381	3,131
Loss and credit carryforwards	1,741	1,441
Amortization	4,159	9,440
Leasing liabilities	6,504	5,041
Unearned revenue	3,717	3,296
Book/tax basis differences in investments and debt	9	373
Capitalized research and development	11,442	6,958
Other	426	489
Deferred income tax assets	33,144	30,850
Less valuation allowance	(1,045)	(939)
Deferred income tax assets, net of valuation allowance	\$ 32,099	\$ 29,911
Deferred Income Tax Liabilities		
Leasing assets	\$ (6,503)	\$ (4,680)
Depreciation	(3,940)	(2,674)
Deferred tax on foreign earnings	(1,837)	(2,738)
Other	(167)	(89)
Deferred income tax liabilities	\$ (12,447)	\$ (10,181)
Net deferred income tax assets	\$ 19,652	\$ 19,730

5. Microsoft reports deferred income tax assets at the end of fiscal years 2024 & 2023 arising from differential book-tax treatment of research and development (R&D). Calculate (a) the *cumulative difference* between amount of R&D expense recognized for tax purposes vs. book purposes at 30/6/2024, and (b) Is the amount of R&D expense recognized during fiscal 2024 for financial reporting purposes smaller or larger than the amount recognized for tax purposes? By how much? You may assume that all R&D expenditures were incurred in the US.

Microsoft: Tax Disclosures

Provision for Income Taxes

The components of the provision for income taxes were as follows:

(In millions)

Year Ended June 30,	2024
Current Taxes	
U.S. federal	\$ 12,165
U.S. state and local	2,366
Foreign	9,858
Current taxes	\$ 24,389
Deferred Taxes	
U.S. federal	\$ (4,791)
U.S. state and local	(379)
Foreign	432
Deferred taxes	\$ (4,738)
Provision for income taxes	\$ 19,651

1. Show the journal entry Microsoft made during fiscal 2024 to provision for income taxes:

Income Tax Expense \$ 19,651

Deferred Taxes (plug) 4,738

Cash/Taxes Payable \$24,389

2. Calculate Microsoft's effective tax rate (ETR) for fiscal 2024. $ETR = 19651/107,787 = 18.2\%$

Provide some reasons why it might be different from the statutory tax rate of 21%?

Permanent book-tax differences, differential tax rates of foreign, state and local earnings

Microsoft's SCF

(In millions)

Year Ended June 30,	2024
Operations	
Net income	\$ 88,136
Adjustments to reconcile net income to net cash from operations:	
Depreciation, amortization, and other	22,287
Stock-based compensation expense	10,734
Net recognized losses (gains) on investments and derivatives	305
Deferred income taxes	(4,738)
Changes in operating assets and liabilities:	
Accounts receivable	(7,191)
Inventories	1,284
Other current assets	(1,648)
Other long-term assets	(6,817)
Accounts payable	3,545
Unearned revenue	5,348
Income taxes	1,687
Other current liabilities	4,867
Other long-term liabilities	749
Net cash from operations	118,548

3. Using the above information, to reconcile the difference between the amount of taxes provisioned during fiscal 2024 vs. the amount of income taxes paid in cash during fiscal 2024.

<i>Income tax Expense</i>	= \$19,651
<i>Change in Deferred Taxes</i>	= 4,738
<i>Taxes Owed</i>	= \$24,389
<i>Change in taxes payable</i>	= 1,687
<i>Cash paid for taxes</i>	\$22,702

<i>Income tax expense</i>	\$ 19,651
<i>Deferred taxes</i>	4,738
<i>Taxes Payable</i>	\$1,687
<i>Cash</i>	22,702

Microsoft: Reconciliation of Statutory vs. Effective Tax Rate

Year Ended June 30,	2024
Federal statutory rate	21.0%
Effect of:	
Foreign earnings taxed at lower rates	(1.4)%
Impact of intangible property transfers	0%
Foreign-derived intangible income deduction	(1.1)%
State income taxes, net of federal benefit	1.5%
Research and development credit	(1.1)%
Excess tax benefits relating to stock-based compensation	(1.1)%
Interest, net	1.1%
Other reconciling items, net	(0.7)%
Effective rate	18.2%

4. The above schedule shows that during fiscal 2024, Microsoft's effective tax rate was higher than its statutory tax rate by 1.1% due the interest, net. What was amount of interest expense that was recognized for GAAP but disallowed for tax purposes during fiscal 2024? You may assume all the interest expense was incurred in the US.

Let the amount of interest disallowed as a tax deduction = X. Then if the effective tax rate increases by 1.1% due the disallowed interest, X must satisfy:

$$0.21 * X / \text{Book income} = 0.21 * X / 107,787 = 1.1\% \Rightarrow X = \$5,645M$$

Microsoft's Schedule of Deferred Tax Items

The components of the deferred income tax assets and liabilities were as follows:

(In millions)		
June 30,	2024	2023
Deferred Income Tax Assets		
Stock-based compensation expense	\$ 765	\$ 681
Accruals, reserves, and other expenses	4,381	3,131
Loss and credit carryforwards	1,741	1,441
Amortization	4,159	9,440
Leasing liabilities	6,504	5,041
Unearned revenue	3,717	3,296
Book/tax basis differences in investments and debt	9	373
Capitalized research and development	11,442	6,958
Other	426	489
Deferred income tax assets	33,144	30,850
Less valuation allowance	(1,045)	(939)
Deferred income tax assets, net of valuation allowance	\$ 32,099	\$ 29,911
Deferred Income Tax Liabilities		
Leasing assets	\$ (6,503)	\$ (4,680)
Depreciation	(3,940)	(2,674)
Deferred tax on foreign earnings	(1,837)	(2,738)
Other	(167)	(89)
Deferred income tax liabilities	\$ (12,447)	\$ (10,181)
Net deferred income tax assets	\$ 19,652	\$ 19,730

Microsoft's Schedule of Deferred Tax Items

5. Microsoft reports deferred income tax assets at the end of fiscal years 2024 & 2023 arising from differential book-tax treatment of research and development (R&D). Calculate (a) the *cumulative difference* between amount of R&D expense recognized for tax purposes vs. book purposes at 30/6/2024, and (b) Is the amount of R&D expense recognized during fiscal 2024 for financial reporting purposes smaller or larger than the amount recognized for tax purposes? By how much?

5(a) Note that the DTA at the end of any fiscal year = (Cumulative GAAP R&D expenses – Cumulative tax-deductible R&D) * statutory tax rate

At the end of fiscal 2024, $DTA_{R\&D} = \$11,442$ million so this implies that cumulative GAAP R&D expenses exceeded cumulative tax-deductible R&D expenses by $\$11,442 / 0.21 = \$54,485$ million.

5(b) Note that $DTA_{R\&D}$ increased by $\$11,442M - 6,958M = \$4,484M$ during fiscal 2024. This implies that during fiscal 2024, R&D expenses recognized for GAAP purposes exceeded R&D expenses recognized for tax purposes by: $\$4,484 / 0.21 = \$21,352$ million. From the income statement, GAAP R&D = \$29,510 million. This implies that R&D recognized on Microsoft's tax return during fiscal 2024 was $\$29,510 - 21,352 = \$8,157$ million.